

Spontaneity, ΔG

ΔG predicts the spontaneity of reaction ; $\Delta G = \Delta H - T\Delta S$

$\Delta G < 0$: spontaneous; $\Delta G > 0$: not spontaneous

Case 1: $\Delta H < 0, \Delta S > 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G =$ - at <u>high</u> temperature, $\Delta G =$
Case 2: $\Delta H > 0, \Delta S < 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G =$ - at <u>high</u> temperature, $\Delta G =$
Case 3: $\Delta H > 0, \Delta S > 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G =$ - at <u>high</u> temperature, $\Delta G =$ ➤ at sufficiently high temperatures, the _____ outweighs _____, hence ΔG is _____
Case 4: $\Delta H < 0, \Delta S < 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G =$ - at <u>high</u> temperature, $\Delta G =$ ➤ at sufficiently high temperatures, the _____ outweighs _____, hence ΔG is _____

Notes for calculating ΔG :

- unit conversions: ΔH is often given in kJ mol^{-1} , ΔS often in $\text{J K}^{-1} \text{mol}^{-1}$.
- T must be in Kelvins.
- ΔG , ΔS and ΔH can be solved by drawing energy cycles.

Notes for $\Delta G = 0$:

- $\Delta G = 0$ for phase changes at its specific temperature
e.g.: $\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(g)}$ at 100°C ; $\Delta G = 0$
e.g.: $\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(s)}$ at 0°C ; $\Delta G = 0$

Entropy, ΔS

Entropy is a measure of the disorder in the system

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

$\Delta S > 0$: _____ in disorder in the system, _____ in entropy

$\Delta S < 0$: _____ in disorder in the system, _____ in entropy

System	ΔS & explanation
1 mol of $\text{H}_2\text{O(s)}$ melts to form $\text{H}_2\text{O(l)}$	- liquid particles _____ - there is an increase in number of ways to arrange the particles and their energies - entropy increases, $\Delta S > 0$
temperature of a sealed vessel containing 1 mol of $\text{N}_2(\text{g})$ is raised from 300 K to 500 K	- broadening of maxwell Boltzmann distribution curve - there is an increase in _____ _____ - entropy increases, $\Delta S > 0$
2 mol of $\text{SO}_2(\text{g})$ completely reacts with 1 mol of $\text{O}_2(\text{g})$ $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$	- decrease in _____ - there is a decrease in _____ _____ - entropy _____

Feasibility

- _____ feasible means $\Delta G < 0$
- _____ feasible means E_a is sufficiently low for reaction to proceed
- _____ favorable means $\Delta H < 0$

Notes for limitation of ΔG to predict whether reaction occurs:

- ΔG gives no information about the rate of reaction (kinetic feasibility)
If $\Delta G < 0$, but activation energy is high, reaction will not proceed at an observable rate

Spontaneity, ΔG

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$\Delta G < 0$: spontaneous; $\Delta G > 0$: not spontaneous

Case 1: $\Delta H < 0, \Delta S > 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G = \text{—ve}$ - at <u>high</u> temperature, $\Delta G = \text{—ve}$
Case 2: $\Delta H > 0, \Delta S < 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G = \text{+ve}$ - at <u>high</u> temperature, $\Delta G = \text{+ve}$
Case 3: $\Delta H > 0, \Delta S > 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G = \text{+ve}$ - at <u>high</u> temperature, $\Delta G = \text{—ve}$ ➤ at sufficiently high temperatures, the <u>negative $-T\Delta S$</u> outweighs <u>positive ΔH</u>, hence ΔG is <u>negative</u>
Case 4: $\Delta H < 0, \Delta S < 0$	$\Delta G = \Delta H - T\Delta S$ - at <u>low</u> temperature, $\Delta G = \text{—ve}$ - at <u>high</u> temperature, $\Delta G = \text{+ve}$ ➤ at sufficiently high temperatures, the <u>positive $-T\Delta S$</u> outweighs <u>negative ΔH</u>, hence ΔG is <u>positive</u>

Notes for calculating ΔG :

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- T must be in Kelvins.
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Entropy, ΔS

Entropy is a measure of the disorder in the system

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

$\Delta S > 0$: increase in disorder in the system, increase in entropy

$\Delta S < 0$: decrease in disorder in the system, decrease in entropy

System	ΔS & explanation
1 mol of $\text{H}_2\text{O(s)}$ melts to form $\text{H}_2\text{O(l)}$	- liquid particles <u>moves around more freely</u> - there is an increase in number of ways to arrange the particles and their energies - entropy increases, $\Delta S > 0$
temperature of a sealed vessel containing 1 mol of $\text{N}_2(\text{g})$ is raised from 300 K to 500 K	- broadening of maxwell Boltzmann distribution curve - there is an increase in <u>number of ways to arrange the energies of the particles</u> - entropy increases, $\Delta S > 0$
2 mol of $\text{SO}_2(\text{g})$ completely reacts with 1 mol of $\text{O}_2(\text{g})$ $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$	- decrease in <u>number of mol of gaseous particles</u> - there is a decrease in <u>number of ways to arrange the particles and their energies</u> - entropy <u>decreases, $\Delta S < 0$</u>

Feasibility

- thermodynamically feasible means $\Delta G < 0$
- kinetically feasible means E_a is sufficiently low for reaction to proceed
- energetically favorable means $\Delta H < 0$

Notes for limitation of ΔG to predict whether reaction occurs:

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